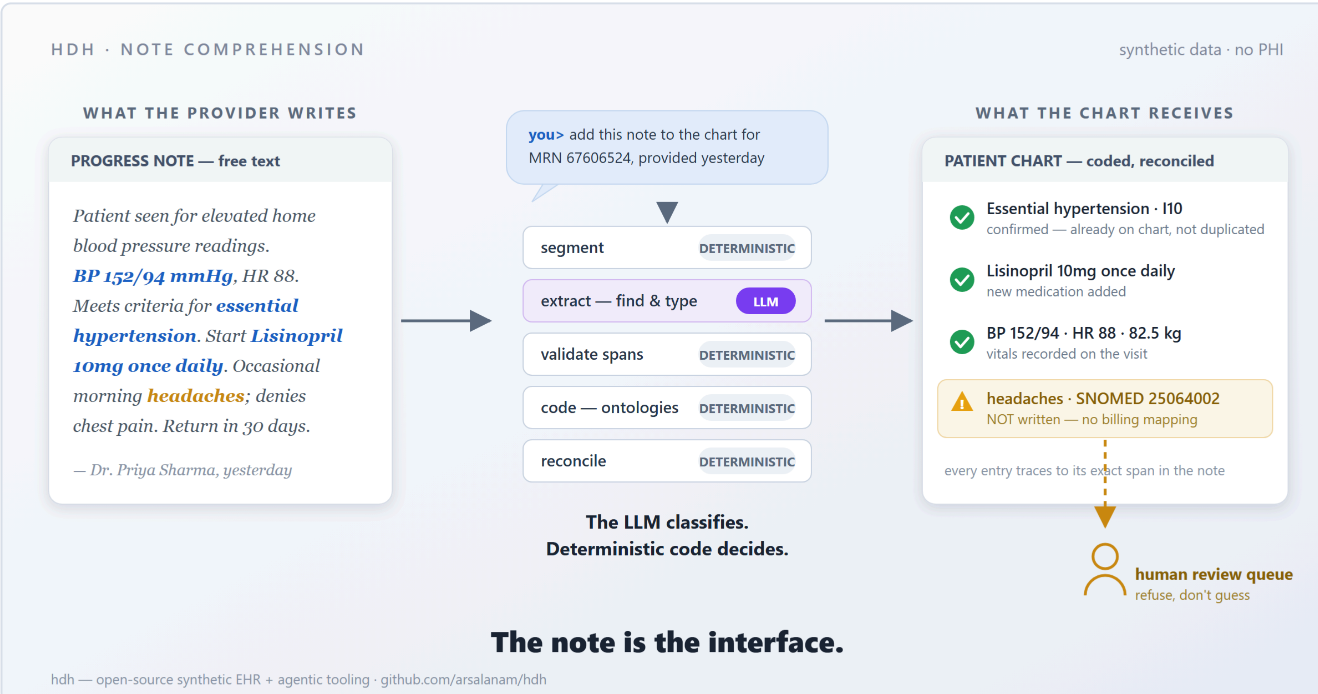
Note Comprehension, or: The Agent Tier Is the New EHR UI



An introduction to the doctor-note comprehension module in hdh, and the roadmap it opens. Everything below runs on synthetic data — hdh generates medically realistic family-medicine charts with zero PHI.

A provider talks to the chart

Here is the whole demo. A provider opens a chat and types:

> *Can you add the following note to the patient chart for patient MRN67606524, provided yesterday by Dr. Priya Sharma: Patient seen in clinic for evaluation of elevated blood pressure readings taken at home over the past month. She reports occasional headaches in the mornings but denies chest pain… blood pressure is 152/94 mmHg… she meets criteria for essential hypertension… Start Lisinopril 10mg once daily for hypertension. Return in 30 days…*

Seconds later, the agent answers with a reconciliation report:

- **Essential hypertension (I10)** — already on the problem list; referenced, not duplicated.
- **Lisinopril 10mg once daily** — new medication added.
- **Vitals** — BP 152/94, HR 88, weight 82.5 kg — recorded.
- ⚠️ **"headaches"** — recognized and coded to SNOMED CT, but no billing mapping exists; **not written**, queued for human review.

No forms. No dropdowns. No twelve-click encounter workflow. The note itself — the thing clinicians actually produce — became the structured, coded, billable chart update. And the one item the system wasn't sure about was *refused*, not guessed.

The design rule that makes it safe

Large language models are superb at reading clinical prose and terrible at being audited. So the pipeline is built on one house rule:

The LLM classifies. Deterministic code decides.

The model does exactly one job: find the clinical mentions in the text and type them (problem, medication, lab/vital, procedure, allergy). It never assigns a code, never decides whether a condition is negated or historical, and never touches the database. Everything after extraction is deterministic, testable code:

- **Verbatim grounding.** Every extracted mention must match the note text character-for-character at a specific span. A mention that can't prove where it came from is rejected, and the model retries with precise feedback. No fuzzy repair, ever.
- **Closed schemas.** Five mention types, a fixed table of attribute kinds, one relation kind. The model cannot invent a category; a new kind is a reviewed schema change, not a free string.
- **Ontology-grounded coding.** Codes come from a deterministic funnel over real terminologies — SNOMED CT for problems and procedures, LOINC for vitals and labs, the drug catalog for medications, ICD-10-CM for billing via curated cross-ontology mappings. The model never "recalls" a code from training data.
- **Rules-first context.** Negation ("denies chest pain"), history, family history, and uncertainty are decided by deterministic rules over the note's sections and trigger words — so "denies chest pain" is provably skipped, not probabilistically skipped.
- **Refuse, don't guess.** Anything that can't be resolved with confidence — an unmapped symptom, a low-confidence link — goes to a human review queue and is never written to the

chart. The reconciler reports four verdicts per entity: `new`, `confirmed`, `review`, `skipped`.

During live testing, the agent once tried to work around a review item by writing raw SQL into the database. The read-only guardrail refused it, the review item stayed unwritten, and the session survived. That is the point: the boundaries hold even when the agent is being creative.

Why this matters: two consumers, one chart

An EHR chart has always had two audiences. Humans read narratives; automated systems — billing, quality measures, decision support, registries, and now AI agents — need codes and structure. Traditional EHRs resolve this tension by making the *clinician* do the structuring, one click at a time, which is how we got the documentation-burden crisis.

Note comprehension inverts it. The clinician produces what they were always going to produce — a note — and the system derives the structure, with provenance: every coded entry points back to the exact span of text it came from, the original note is stored on the visit verbatim, and the same visit round-trips back out as a SOAP note or a FHIR document bundle. The chart stays usable by humans, by AI, and by the decidedly non-AI automation that actually runs healthcare.

The thesis: the agent tier is the new UI

Once free text can safely become structured chart data, the screen full of forms stops being the interface. The interface is a conversation backed by **tools with contracts**:

- the agent holds *published, guarded tools* — chart a note, query a cohort by ontology subtree, normalize a term, look up a code — each one a narrow API that validates its inputs and rolls back on failure;
- the ontologies are the shared vocabulary between the human, the agent, and every downstream system;
- the guardrails and review queues are where clinical accountability lives — the human is *in* the loop by design, not by exception.

In that world the EHR's "UI tier" is an agent tier. Screens become one optional client among many; the real product is the governed set of capabilities the agent can exercise on the chart.

The roadmap

The comprehension module is merged and live-tested in hdh today (design docs: the service and the extraction contract). What comes next is the rest of the chart-maintenance surface:

1. **Comprehensive testing & eval discipline** — a replay corpus where every live model failure becomes a permanent regression fixture, plus a measured baseline (recall, precision, linking, assertion accuracy) that no future prompt or model change may silently regress.
2. **Amend / delete + an append-only audit log** — the sanctioned edit path. When a charted item is wrong, the provider tells the agent to fix it; the audit trail records who changed what, when, and why. This is what makes agent-maintained charts *credible*, not just possible.
3. **Broader billing coverage** — curated symptom mappings so common complaints (headache, fatigue, dizziness) chart automatically instead of queueing for review, while unmapped items keep refusing to guess.
4. **Orders through the agent** — the same comprehension of "basic metabolic panel to be drawn before the next visit" becoming a real service request, with the same verdict-and-review discipline.
5. **Care plans** — concerns linked to interventions, goals tracked across visits: the structure comprehension extracts today becoming the substrate a longitudinal care-plan agent reasons over.

Each step follows the same pattern: a narrow, contract-first capability; deterministic code wherever a decision has consequences; the LLM only where language understanding is genuinely the job; and a human review path wherever confidence runs out.

Try it

hdh is open source (MIT): github.com/arsalanam/hdh. Generate a synthetic panel, load the ontologies, and paste a note at the agent — the README's *"Chart free-text notes through the agent"* section has the five-minute version. Feedback, issues, and skepticism are all welcome; the review queue was built for exactly that.